

Rainul Hakim

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EDUCATION

Davidson College

Bachelor of Science, Computer Science

Davidson, NC

Expected May 2029

- Relevant Coursework: Data Structures, Programming and Problem Solving, Database Programming, Linear Algebra, Calculus I & Modeling, Calculus II

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL

ML / Scientific Computing: PyTorch, PyTorch Geometric, TensorFlow.js, MediaPipe, NumPy, scikit-learn

Web: React, Node.js, Vercel Serverless Functions, HTML, CSS

Tools: Git, GitHub, VS Code, HPC clusters

EXPERIENCE

Machine Learning Research Assistant, ALPhA Lab, Davidson College

Summer 2026

In collaboration with the Facility for Rare Isotope Beams (FRIB), Michigan State University

- Applied contrastive self-supervised learning with a DGCNN encoder to raw 3D point cloud data from the AT-TPC detector, learning representations of collision events without manual labels, toward a foundation model for TPC data adaptable to multiple downstream physics tasks
- Evaluated representations with linear probing and ran systematic augmentation ablations to improve performance
- Trained on real experimental data with PyTorch and PyTorch Geometric on an HPC cluster; contributed reproducible, documented code to a shared cross-institutional codebase

Student Consultant, Data CATS (Consulting, Analytics, Tutoring Services), Davidson College

Aug 2026 - Present

- Consult and tutor students, faculty, and staff on data analytics, from framing questions to analysis and visualization

Student Consultant, The Hurt Hub@Davidson

Aug 2026 - Present

- Advise a client partner on an early-stage startup through Davidson's entrepreneurship hub

PROJECTS

AlternAte - AI Nutrition Assistant

React, Vite, Vercel Serverless, Gemini API | Hackathon

- Built an AI web app that analyzes meals and returns macro breakdowns, health flags, and goal-based recommendations
- Secured the Gemini API behind serverless functions with structured JSON validation; added a meal comparison engine and weekly scoring system with client-side persistence

In-Browser Hand Gesture Recognition (ML)

JavaScript, TensorFlow.js, MediaPipe | Open-source extension

- Built a gesture-triggered effects app running fully in-browser at ~30 fps with no backend: MediaPipe Holistic extracts 126 hand-landmark coordinates per frame and feeds a TensorFlow.js neural network
- Normalized landmarks by wrist position and scale for hand-size invariance; trained a binary classifier (Dropout 0.3, 50 epochs, 0.97 confidence threshold) with a custom in-browser training UI
- Segmented the user's silhouette with MediaPipe Selfie Segmentation and rendered multiple clones with smoke effects on Canvas

Left No Crumbs - 1st Place (\$1,000), Ideasprint

Startup concept

- Co-developed a campus food-waste startup with a real-time notification workflow for surplus catered meals; built the pitch deck, business model, and technical roadmap in 4 weeks and beat competing student teams

LEADERSHIP & INVOLVEMENT

Resident Advisor, Davidson College

Aug 2026 - Present

- Support 30+ residents via community-building programming, conflict resolution, and crisis response

Event Coordinator, Davidson Entrepreneurship Club

Nov 2025 - Present

- Coordinate and promote innovation-focused events for student founders; design visual content for outreach

Secretary, Association for Computing Machinery (ACM) at Davidson College

Treasurer, International Association, Davidson College

Co-Founder, Project BeWell, Chattogram, Bangladesh

Aug 2023 - Sep 2023

- Led a 12-member team on a public health initiative reaching 1,000+ people; secured 10K+ BDT in funding; ran medical camps with 99% satisfaction